

## Qualification Pack



# Automotive Battery Management System (BMS) Design Engineer

QP Code: ASC/Q8315

Version: 1.0

NSQF Level: 5.5

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## Qualification Pack

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## Qualification Pack

# ASC/Q8315: Automotive Battery Management System (BMS) Design Engineer

## Brief Job Description

Individual at this job should benchmark, understand, release design and architecture of BMS system & its components for Electric Vehicle (EV) as per the specified norms and standards. BMS engineer also supports manager and core team during implementation and field issue resolution after implementation

## Personal Attributes

He/she must have the ability to plan, analyse and assess the solutions co-work with stakeholders, have clarity and clear understanding of safety, recycle and reuse standards and latest government Policies and regulations, work in a team, good communication skills, tech-savvy individual as well.

## Applicable National Occupational Standards (NOS)

### Compulsory NOS:

1. [ASC/N9810: Manage work and resources \(Manufacturing\)](#)
2. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)
3. [ASC/N8335: Review the targeted design architecture of EV](#)
4. [ASC/N8336: Develop prominent options of BMS architecture, infrastructure and solutions](#)
5. [ASC/N8337: Conduct simulation for verification & validate of various architectures](#)
6. [ASC/N8338: Support manager to execute implementation of BMS system](#)

## Qualification Pack (QP) Parameters

|                   |                                |
|-------------------|--------------------------------|
| <b>Sector</b>     | Automotive                     |
| <b>Sub-Sector</b> | Research & Development         |
| <b>Occupation</b> | Automotive Product Development |
| <b>Country</b>    | India                          |
| <b>NSQF Level</b> | 5.5                            |
| <b>Credits</b>    | 21                             |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/8212.0100, 2015/8212.0200                                                                                                                                                                                                                                                                                                                 |
| <b>Minimum Educational Qualification &amp; Experience</b> | Completed 3 year diploma after 10th (Mechanical/Automobile/ Electrical / Electronics) with 3 Years of experience in relevant field<br>OR<br>B.E. (Pursuing 4th year)<br>OR<br>Certificate-NSQF (Electric Vehicle Product Design Engineer/ Automotive Prototype Manufacturing Lead Technician Level 5) with 2 Years of experience in relevant field |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                                                                    |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                 |
| <b>Minimum Job Entry Age</b>                              | 22 Years                                                                                                                                                                                                                                                                                                                                           |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                 |
| <b>Next Review Date</b>                                   | 28/02/2026                                                                                                                                                                                                                                                                                                                                         |
| <b>NSQC Approval Date</b>                                 | 28/02/2023                                                                                                                                                                                                                                                                                                                                         |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                |
| <b>Reference code on NQR</b>                              | QG-5.5-AU-00227-2023-V1-ASDC                                                                                                                                                                                                                                                                                                                       |
| <b>NQR Version</b>                                        | 1.1                                                                                                                                                                                                                                                                                                                                                |

## Qualification Pack

### ASC/N9810: Manage work and resources (Manufacturing)

#### Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

#### Scope

The scope covers the following :

- Maintain safe and secure working environment
- Maintain Health and Hygiene
- Effective waste management practices
- Material/energy conservation practices

#### Elements and Performance Criteria

##### *Maintain safe and secure working environment*

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** implement safe working practices for dealing with hazards to ensure safety of self and others
- PC3.** conduct regular checks of the machines with support of the maintenance team to identify potential hazards
- PC4.** ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions
- PC5.** organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices
- PC6.** fill daily check sheet to report improvements done and risks identified
- PC7.** ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others
- PC8.** report any identified breaches in health, safety and security policies and procedures to the designated person

##### *Maintain Health and Hygiene*

To be competent, the user/individual on the job must be able to:

- PC9.** ensure workplace, equipment, restrooms etc. are sanitized regularly
- PC10.** ensure team is aware about hygiene and sanitation regulations and following them on the shop floor
- PC11.** ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace
- PC12.** report advanced hygiene and sanitation issues to appropriate authority
- PC13.** follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc
- PC14.** wear and dispose PPEs regularly and appropriately

## Qualification Pack

### *Effective waste management practices*

To be competent, the user/individual on the job must be able to:

**PC15.** ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP

**PC16.** ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste

### *Material/energy conservation practices*

To be competent, the user/individual on the job must be able to:

**PC17.** ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively

**PC18.** prepare and analyze material and energy audit reports to decipher excessive consumption of material and water

**PC19.** identify possibilities of using renewable energy and environment friendly fuels

**PC20.** identify processes where material and energy/electricity utilization can be optimized

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context

**KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same

**KU3.** evacuation procedures for workers and visitors

**KU4.** how and when to report hazards as well as the limits of responsibility for dealing with hazards

**KU5.** potential hazards, risks and threats based on the nature of work

**KU6.** various types of fire extinguisher

**KU7.** various types of safety signs and their meaning

**KU8.** appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

**KU9.** relevant standards, procedures and policies related to 5S followed in the company

**KU10.** the various materials used and their storage norms

**KU11.** importance of efficient utilisation of material and water

**KU12.** basics of electricity and prevalent energy efficient devices

**KU13.** common practices of conserving electricity

**KU14.** common sources and ways to minimize pollution

**KU15.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

**KU16.** waste management techniques

**KU17.** significance of greening

## Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Maintain safe and secure working environment</i>                                                                                                                                                       | <b>20</b>    | <b>13</b>       | -             | <b>8</b>   |
| <b>PC1.</b> identify hazardous activities and the possible causes of risks or accidents in the workplace                                                                                                  | 4            | 2               | -             | 2          |
| <b>PC2.</b> implement safe working practices for dealing with hazards to ensure safety of self and others                                                                                                 | 3            | 1               | -             | 2          |
| <b>PC3.</b> conduct regular checks of the machines with support of the maintenance team to identify potential hazards                                                                                     | 2            | 2               | -             | 1          |
| <b>PC4.</b> ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions | 3            | 2               | -             | 1          |
| <b>PC5.</b> organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices                                                                   | 2            | -               | -             | -          |
| <b>PC6.</b> fill daily check sheet to report improvements done and risks identified                                                                                                                       | 2            | 2               | -             | -          |
| <b>PC7.</b> ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others                                                                                       | 2            | 2               | -             | 1          |
| <b>PC8.</b> report any identified breaches in health, safety and security policies and procedures to the designated person                                                                                | 2            | 2               | -             | 1          |
| <i>Maintain Health and Hygiene</i>                                                                                                                                                                        | <b>13</b>    | <b>7</b>        | -             | <b>5</b>   |
| <b>PC9.</b> ensure workplace, equipment, restrooms etc. are sanitized regularly                                                                                                                           | 3            | 2               | -             | 1          |
| <b>PC10.</b> ensure team is aware about hygiene and sanitation regulations and following them on the shop floor                                                                                           | 2            | 1               | -             | -          |
| <b>PC11.</b> ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace                                                                                                | 2            | 2               | -             | 1          |



### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> report advanced hygiene and sanitation issues to appropriate authority                                                       | 1            | 1               | -             | 1          |
| <b>PC13.</b> follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc                       | 2            | 1               | -             | 1          |
| <b>PC14.</b> wear and dispose PPEs regularly and appropriately                                                                            | 3            | -               | -             | 1          |
| <i>Effective waste management practices</i>                                                                                               | <b>6</b>     | <b>4</b>        | -             | <b>1</b>   |
| <b>PC15.</b> ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP                                             | 3            | 2               | -             | -          |
| <b>PC16.</b> ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste          | 3            | 2               | -             | 1          |
| <i>Material/energy conservation practices</i>                                                                                             | <b>11</b>    | <b>6</b>        | -             | <b>6</b>   |
| <b>PC17.</b> ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively | 2            | 2               | -             | 1          |
| <b>PC18.</b> prepare and analyze material and energy audit reports to decipher excessive consumption of material and water                | 3            | 2               | -             | 1          |
| <b>PC19.</b> identify possibilities of using renewable energy and environment friendly fuels                                              | 3            | 1               | -             | 2          |
| <b>PC20.</b> identify processes where material and energy/electricity utilization can be optimized                                        | 3            | 1               | -             | 2          |
| <b>NOS Total</b>                                                                                                                          | <b>50</b>    | <b>30</b>       | -             | <b>20</b>  |

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### National Occupational Standards (NOS) Parameters

|                            |                                           |
|----------------------------|-------------------------------------------|
| <b>NOS Code</b>            | ASC/N9810                                 |
| <b>NOS Name</b>            | Manage work and resources (Manufacturing) |
| <b>Sector</b>              | Automotive                                |
| <b>Sub-Sector</b>          | Generic                                   |
| <b>Occupation</b>          | Generic                                   |
| <b>NSQF Level</b>          | 5                                         |
| <b>Credits</b>             | 2                                         |
| <b>Version</b>             | 2.0                                       |
| <b>Last Reviewed Date</b>  | 08/05/2025                                |
| <b>Next Review Date</b>    | 08/05/2028                                |
| <b>NSQC Clearance Date</b> | 08/05/2025                                |

## Qualification Pack

### DGT/VSQ/N0103: Employability Skills (90 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

##### *Constitutional values - Citizenship*

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the current job market requirement and future of work                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability relevant portals                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC3.</b> research about the different industries, job market trends, latest skills required and the available opportunities                                                                                                                                                                                   | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC4.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC5.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC6.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC7.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <b>PC8.</b> adopt a continuous learning mindset for personal and professional development                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |

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| Assessment Criteria for Outcomes                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                      | -            | -               | -             | -          |
| <b>PC11.</b> write short messages, notes, letters, e-mails etc. in English                                                                                             | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                           | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> identify career goals based on the skills, interests, knowledge, and personal attributes                                                                  | -            | -               | -             | -          |
| <b>PC13.</b> prepare a career development plan with short- and long-term goals                                                                                         | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                            | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> follow verbal and non-verbal communication etiquette while communicating in professional and public settings                                              | -            | -               | -             | -          |
| <b>PC15.</b> use active listening techniques for effective communication                                                                                               | -            | -               | -             | -          |
| <b>PC16.</b> communicate in writing using appropriate style and format based on formal or informal requirements                                                        | -            | -               | -             | -          |
| <b>PC17.</b> work collaboratively with others in a team                                                                                                                | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC18.</b> communicate and behave appropriately with all genders and PwD                                                                                             | -            | -               | -             | -          |
| <b>PC19.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                                                                       | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC20.</b> identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc. | -            | -               | -             | -          |
| <b>PC21.</b> carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook                     | -            | -               | -             | -          |



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| Assessment Criteria for Outcomes                                                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC22.</b> identify common components of salary and compute income, expenses, taxes, investments etc                                                                               | -            | -               | -             | -          |
| <b>PC23.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                                                                                | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                                                                                      | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC24.</b> operate digital devices and use their features and applications securely and safely                                                                                     | -            | -               | -             | -          |
| <b>PC25.</b> carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc. | -            | -               | -             | -          |
| <b>PC26.</b> display responsible online behaviour while using various social media platforms                                                                                         | -            | -               | -             | -          |
| <b>PC27.</b> create a personal email account, send and process received messages as per requirement                                                                                  | -            | -               | -             | -          |
| <b>PC28.</b> carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications                                                   | -            | -               | -             | -          |
| <b>PC29.</b> utilize virtual collaboration tools to work effectively                                                                                                                 | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                                                                                                              | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC30.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                           | -            | -               | -             | -          |
| <b>PC31.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                          | -            | -               | -             | -          |
| <b>PC32.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                               | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                              | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC33.</b> identify different types of customers and ways to communicate with them                                                                                                 | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC34.</b> identify and respond to customer requests and needs in a professional manner                                                                                        | -            | -               | -             | -          |
| <b>PC35.</b> use appropriate tools to collect customer feedback                                                                                                                  | -            | -               | -             | -          |
| <b>PC36.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC37.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC38.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC39.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC40.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC41.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0103                   |
| <b>NOS Name</b>            | Employability Skills (90 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 5                               |
| <b>Credits</b>             | 3                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 07/10/2025                      |
| <b>Next Review Date</b>    | 07/10/2028                      |
| <b>NSQC Clearance Date</b> | 07/10/2025                      |

## Qualification Pack

### ASC/N8335: Review the targeted design architecture of EV

#### Description

This NOS unit is about performing tasks related to review target vehicle design architecture, its various systems, Battery pack design, applications & customer usage requirements.

#### Scope

The scope covers the following :

- Assess the requirements
- Assessing battery system, cell design & its Thermal Management System design

#### Elements and Performance Criteria

##### *Assess the requirements*

To be competent, the user/individual on the job must be able to:

- PC1.** Evaluate the type of EV that is 2W/3W/4W for information about vehicle design architecture, its components, and operational parameters to be received from the customer/OEM for the BMS infrastructure
- PC2.** Obtain the benchmarking data of previous project and evaluate the targeted applications and market requirements for the current project
- PC3.** Check the types of BSS solution to be deployed to meet the customer requirement
- PC4.** Identify prominent E/E issues & current leakages in the architecture
- PC5.** Identify required hardware & software for BMS with estimated budgeting
- PC6.** Propose the best suitable solutions to manager for the selection

##### *Assessing battery system, cell design & its Thermal Management System design*

To be competent, the user/individual on the job must be able to:

- PC7.** Prepare the outline for proposed battery swapping system and cell design packaging
- PC8.** Evaluate proposed battery thermal management system design considering applications and vehicle architecture
- PC9.** Evaluate E component packaging space & location to check for temperature & safety
- PC10.** Review several market solutions and latest trends for the similar applications and best practices to carefully choose from available best solutions
- PC11.** Benchmark BMS in the market comparing target vehicle
- PC12.** Review SOC, SOH, Cell Chemistry & cell design, Safety measurements for similar applications

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation policies, procedures, and guidelines that relate to designing and maintaining networks
- KU2.** market trends of latest technologies, types of EVs, types of EV Batteries

## Qualification Pack

- KU3.** E/E drawings & vehicle layout architectures
- KU4.** BMS System & EV basics, its components & working principals
- KU5.** range of standard templates and tools available and how to use them
- KU6.** process layouts, drawings & other technical details
- KU7.** energy consumptions & usage
- KU8.** energy controlling & monitoring systems, its types & limitations
- KU9.** updated internal and external regulations for systems design
- KU10.** impact of organisational processes & products on the environment and human health safety guidelines

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Assess the requirements</i>                                                                                                                                                                                     | <b>20</b>    | <b>20</b>       | -             | <b>10</b>  |
| <b>PC1.</b> Evaluate the type of EV that is 2W/3W/4W for information about vehicle design architecture, its components, and operational parameters to be received from the customer/OEM for the BMS infrastructure | 4            | 4               | -             | 3          |
| <b>PC2.</b> Obtain the benchmarking data of previous project and evaluate the targeted applications and market requirements for the current projec                                                                 | 3            | 3               | -             | 2          |
| <b>PC3.</b> Check the types of BSS solution to be deployed to meet the customer requirement                                                                                                                        | 3            | 3               | -             | 1          |
| <b>PC4.</b> Identify prominent E/E issues & current leakages in the architecture                                                                                                                                   | 4            | 4               | -             | 1          |
| <b>PC5.</b> Identify required hardware & software for BMS with estimated budgeting                                                                                                                                 | 3            | 3               | -             | 1          |
| <b>PC6.</b> Propose the best suitable solutions to manager for the selection                                                                                                                                       | 3            | 3               | -             | 2          |
| <i>Assessing battery system, cell design &amp; its Thermal Management System design</i>                                                                                                                            | <b>20</b>    | <b>20</b>       | -             | <b>10</b>  |
| <b>PC7.</b> Prepare the outline for proposed battery swapping system and cell design packaging                                                                                                                     | 3            | 3               | -             | 1          |
| <b>PC8.</b> Evaluate proposed battery thermal management system design considering applications and vehicle architecture                                                                                           | 4            | 4               | -             | 2          |
| <b>PC9.</b> Evaluate E component packaging space & location to check for temperature & safety                                                                                                                      | 4            | 4               | -             | 2          |
| <b>PC10.</b> Review several market solutions and latest trends for the similar applications and best practices to carefully choose from available best solutions                                                   | 4            | 4               | -             | 2          |
| <b>PC11.</b> Benchmark BMS in the market comparing target vehicle                                                                                                                                                  | 3            | 3               | -             | 2          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> Review SOC, SOH, Cell Chemistry & cell design, Safety measurements for similar applications | 2            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                         | <b>40</b>    | <b>40</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                               |
|----------------------------|-----------------------------------------------|
| <b>NOS Code</b>            | ASC/N8335                                     |
| <b>NOS Name</b>            | Review the targeted design architecture of EV |
| <b>Sector</b>              | Automotive                                    |
| <b>Sub-Sector</b>          | Research & Development                        |
| <b>Occupation</b>          | Automotive Product Development                |
| <b>NSQF Level</b>          | 5.5                                           |
| <b>Credits</b>             | 3                                             |
| <b>Version</b>             | 1.0                                           |
| <b>Last Reviewed Date</b>  | 28/02/2023                                    |
| <b>Next Review Date</b>    | 28/02/2026                                    |
| <b>NSQC Clearance Date</b> | 28/02/2023                                    |



## Qualification Pack

### ASC/N8336: Develop prominent options of BMS architecture, infrastructure and solutions

#### Description

This NOS unit is about performing task related to defining & prepare various concepts for BMS systems, and choose most prominent options based on design requirements. This also include design of Hardware. Software, UI/UX & Control System, its parameters, safety structure, operational parameters, SOC, SOH, cell balancing & battery life estimations, budgetary requirements.

#### Scope

The scope covers the following :

- Prepare hardware, software & Control system components required for BMS

#### Elements and Performance Criteria

##### *Prepare hardware, software & Control system components required for BMS*

To be competent, the user/individual on the job must be able to:

- PC1.** Analyse & initiate concept for BMS, Hardware & Software configurations
- PC2.** Follow government norms for EV, legal requirements, safety requirements per ISO 26262 & compliances for designing BMS
- PC3.** Conduct core team interactions to define boundaries for BMS architectures
- PC4.** Prepare hardware requirement for BMS system configurations along with ECU & Sensor integration mapping)
- PC5.** Define / Develop Software architecture using MBD approach
- PC6.** Define control system algorithms using Coulomb Counting Method or Kalman Filter Method for estimating SOC, SOH, SOP, SOS, Fault Detection code & diagnostics, Battery Life Estimation, Charging & Discharging monitoring & Controlling mechanisms
- PC7.** Prepare estimated budget requirements
- PC8.** Submit the same for approval to the management

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Product portfolio of organization
- KU2.** Basics of SOH, SOC, SOP, SOS
- KU3.** Standard Operation Procedures (SOP) recommended by manufacturer for using different components
- KU4.** MBD approach & Simulation software like Simulink, MATLAB
- KU5.** Batter designs, cell design, cell balancing , Types of Microcontrollers, PCB Design
- KU6.** Impact of Thermal & external parameters on performance of BMS systems

## Qualification Pack

- KU7.** Thermal Management of Battery System & its requirements
- KU8.** Different types of Sensors, Integrations Policies, CAN Bus, J1939, J1772 Protocol
- KU9.** Communication data protocols referring ISO 15118, OCPP, OCPI and other protocols
- KU10.** Types of Charging Stations, connector types & protocols
- KU11.** Government Norms, Legal Requirements, Safety requirements as per ISO 26262 or similar & international compliance requirements
- KU12.** using system design software like Simulia, Ansys Medini Analyse, SCADA, etc

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare hardware, software &amp; Control system components required for BMS</i>                                                                                                                                                                            | <b>40</b>    | <b>40</b>       | -             | <b>20</b>  |
| <b>PC1.</b> Analyse & initiate concept for BMS, Hardware & Software configurations                                                                                                                                                                            | 5            | 5               | -             | 3          |
| <b>PC2.</b> Follow government norms for EV, legal requirements, safety requirements per ISO 26262 & compliances for designing BMS                                                                                                                             | 5            | 5               | -             | 2          |
| <b>PC3.</b> Conduct core team interactions to define boundaries for BMS architectures                                                                                                                                                                         | 6            | 6               | -             | 2          |
| <b>PC4.</b> Prepare hardware requirement for BMS system configurations along with ECU & Sensor integration mapping)                                                                                                                                           | 5            | 5               | -             | 3          |
| <b>PC5.</b> Define / Develop Software architecture using MBD approach                                                                                                                                                                                         | 6            | 6               | -             | 3          |
| <b>PC6.</b> Define control system algorithms using Coulomb Counting Method or Kalman Filter Method for estimating SOC, SOH, SOP, SOS, Fault Detection code & diagnostics, Battery Life Estimation, Charging & Discharging monitoring & Controlling mechanisms | 7            | 7               | -             | 4          |
| <b>PC7.</b> Prepare estimated budget requirements                                                                                                                                                                                                             | 4            | 4               | -             | 2          |
| <b>PC8.</b> Submit the same for approval to the management                                                                                                                                                                                                    | 2            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                                                                                                              | <b>40</b>    | <b>40</b>       | -             | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                             |
|----------------------------|-----------------------------------------------------------------------------|
| <b>NOS Code</b>            | ASC/N8336                                                                   |
| <b>NOS Name</b>            | Develop prominent options of BMS architecture, infrastructure and solutions |
| <b>Sector</b>              | Automotive                                                                  |
| <b>Sub-Sector</b>          | Research & Development                                                      |
| <b>Occupation</b>          | Automotive Product Development                                              |
| <b>NSQF Level</b>          | 5.5                                                                         |
| <b>Credits</b>             | 3                                                                           |
| <b>Version</b>             | 1.0                                                                         |
| <b>Last Reviewed Date</b>  | 28/02/2023                                                                  |
| <b>Next Review Date</b>    | 28/02/2026                                                                  |
| <b>NSQC Clearance Date</b> | 28/02/2023                                                                  |

## Qualification Pack

### ASC/N8337: Conduct simulation for verification & validate of various architectures

#### Description

This NOS unit is about performing simulations using software, planning various performance, durability & safety testing for BMS system, support testing & validation activities, comparing results and proposing most suitable BMS system for target vehicle application.

#### Scope

The scope covers the following :

- Verify design options and plan for Design validation and planning (DVP) requirements
- Publish technical verification / validations results and architectures options

#### Elements and Performance Criteria

##### *Verify design options and plan for Design validation and planning (DVP) requirements*

To be competent, the user/individual on the job must be able to:

- PC1.** Complete design simulations using software like MATLAB or Simulink reflecting vehicle applications, loading conditions, Charging & Discharging Profiles, SOC estimations based on Coulomb counting or Kalman Filter algorithms etc.
- PC2.** Review simulation results for Cell Voltage, Temperature, SOC, SOH, cell balancing
- PC3.** Discuss with superior & perform design improvements for BMS architecture
- PC4.** Benchmark technical guidelines TGR/TGW for similar BMS
- PC5.** Get the design Validation Plan (DVP) considering design requirements, loading conditions, vehicle applications, usage patterns, Governing legal regulations & Safety standards & requirements, ASIC C/D Compliance, EMI/EMC Requirements
- PC6.** Support testing & planning team to complete testing / validations on technical topics

##### *Publish technical verification / validations results and architectures options*

To be competent, the user/individual on the job must be able to:

- PC7.** Prepare comparison study with verification & testing results outcomes
- PC8.** Discuss & prepare most suitable BMS architecture option recommendation considering design parameters, adaption complexity, cost

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Product portfolio of organization
- KU2.** Product Benchmarking & Market technology analysis
- KU3.** Standard Operation Procedures (SOP) recommended by manufacturer for using equipment / machinery in use
- KU4.** Internal Responsible departments and team members

## Qualification Pack

- KU5.** Leading component suppliers and competition designs
- KU6.** Use of Simulation software like MATLAB or Simulink
- KU7.** SOC Estimations using Coulomb counting or Kalman Filter Algorithms
- KU8.** Standard testing requirements & testing procedures, DVP for various components & systems
- KU9.** Limitations of various BMS architectures & its possible resolutions
- KU10.** System Fault codes & its Diagnostics, FMEA methodology, Predictive Cell Diagnosis & Remaining Useful life estimations
- KU11.** Change Management, Stakeholder review, results analysis
- KU12.** Integrated Reporting to all stakeholders

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Verify design options and plan for Design validation and planning (DVP) requirements</i>                                                                                                                                                          | 32           | 32              | -             | 16         |
| <b>PC1.</b> Complete design simulations using software like MATLAB or Simulink reflecting vehicle applications, loading conditions, Charging & Discharging Profiles, SOC estimations based on Coulomb counting or Kalman Filter algorithms etc.      | 7            | 7               | -             | 5          |
| <b>PC2.</b> Review simulation results for Cell Voltage, Temperature, SOC, SOH, cell balancing                                                                                                                                                        | 5            | 5               | -             | 3          |
| <b>PC3.</b> Discuss with superior & perform design improvements for BMS architecture                                                                                                                                                                 | 5            | 5               | -             | 2          |
| <b>PC4.</b> Benchmark technical guidelines TGR/TGW for similar BMS                                                                                                                                                                                   | 4            | 4               | -             | 2          |
| <b>PC5.</b> Get the design Validation Plan (DVP) considering design requirements, loading conditions, vehicle applications, usage patterns, Governing legal regulations & Safety standards & requirements, ASIC C/D Compliance, EMI/EMC Requirements | 7            | 7               | -             | 2          |
| <b>PC6.</b> Support testing & planning team to complete testing / validations on technical topics                                                                                                                                                    | 4            | 4               | -             | 2          |
| <i>Publish technical verification / validations results and architectures options</i>                                                                                                                                                                | 8            | 8               | -             | 4          |
| <b>PC7.</b> Prepare comparison study with verification & testing results outcomes                                                                                                                                                                    | 4            | 4               | -             | 2          |
| <b>PC8.</b> Discuss & prepare most suitable BMS architecture option recommendation considering design parameters, adaption complexity, cost                                                                                                          | 4            | 4               | -             | 2          |
| <b>NOS Total</b>                                                                                                                                                                                                                                     | <b>40</b>    | <b>40</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>NOS Code</b>            | ASC/N8337                                                               |
| <b>NOS Name</b>            | Conduct simulation for verification & validate of various architectures |
| <b>Sector</b>              | Automotive                                                              |
| <b>Sub-Sector</b>          | Research & Development                                                  |
| <b>Occupation</b>          | Automotive Product Development                                          |
| <b>NSQF Level</b>          | 5.5                                                                     |
| <b>Credits</b>             | 3                                                                       |
| <b>Version</b>             | 1.0                                                                     |
| <b>Last Reviewed Date</b>  | 28/02/2023                                                              |
| <b>Next Review Date</b>    | 28/02/2026                                                              |
| <b>NSQC Clearance Date</b> | 28/02/2023                                                              |



## Qualification Pack

### ASC/N8338: Support manager to execute implementation of BMS system

#### Description

This NOS unit is about preparing detail planning for the selected BMS architecture, releasing drawings & specification books, safety requirements, fault codes and diagnosis plan. This is also about providing system implementation support to the core team and manager, field issue resolution support.

#### Scope

The scope covers the following :

- Prepare for implementation of BMS
- Support manager & project teams to execute implementation of BMS

#### Elements and Performance Criteria

##### *Prepare for implementation of BMS*

To be competent, the user/individual on the job must be able to:

- PC1.** Obtain the approval from the management on the selected proposal
- PC2.** Prepare detailed design for H/W & Software Interfaces, UI/UX interfaces
- PC3.** Prepare control system detailed design with defined architecture & strategies
- PC4.** Release detail design, architecture Drawings for development
- PC5.** Prepare & release specification book for various components & system

##### *Support manager & project teams to execute implementation of BMS*

To be competent, the user/individual on the job must be able to:

- PC6.** Participate in design reviews with the team internally & externally
- PC7.** Support the core team for necessary technical clarifications & resolutions from R&D network
- PC8.** Support for vehicle integration & series implementation
- PC9.** Participate quality meetings & receive BMS performance feedback
- PC10.** Understand field issues regarding BMS related to various architecture, durability or control system issues
- PC11.** Propose & provide necessary technical resolution for the issues during deployment

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Vehicle application & Market usage patterns
- KU2.** Electrical connection layout, energy maps, energy costs
- KU3.** Various materials used in BMS and their environmental impacts
- KU4.** Battery power cycles and prevalent energy efficient devices
- KU5.** Types of Batteries used in EV, Loading cycles and its impacts on battery life, Safety requirements for selected battery types

## Qualification Pack

- KU6.** Communicating with internal & external stakeholders
- KU7.** Preparation of Design & Drawings for system and various components, Conventions used in E/E drawings, Product design management & release
- KU8.** EMI/EMC requirements
- KU9.** Updated internal and external regulations for system and component designs
- KU10.** Latest technology discussion forums and future technology study
- KU11.** Understanding Field issues, its relationship with BMS system & its resolutions
- KU12.** Team working & communications
- KU13.** Latest Automotive trends & development strategies

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare for implementation of BMS</i>                                                                                | <b>19</b>    | <b>19</b>       | -             | <b>9</b>   |
| <b>PC1.</b> Obtain the approval from the management on the selected proposal                                            | 3            | 3               | -             | 1          |
| <b>PC2.</b> Prepare detailed design for H/W & Software Interfaces, UI/UX interfaces                                     | 5            | 5               | -             | 2          |
| <b>PC3.</b> Prepare control system detailed design with defined architecture & strategies                               | 4            | 4               | -             | 2          |
| <b>PC4.</b> Release detail design, architecture Drawings for development                                                | 4            | 4               | -             | 2          |
| <b>PC5.</b> Prepare & release specification book for various components & system                                        | 3            | 3               | -             | 2          |
| <i>Support manager &amp; project teams to execute implementation of BMS</i>                                             | <b>21</b>    | <b>21</b>       | -             | <b>11</b>  |
| <b>PC6.</b> Participate in design reviews with the team internally & externally                                         | 4            | 4               | -             | 2          |
| <b>PC7.</b> Support the core team for necessary technical clarifications & resolutions from R&D network                 | 4            | 4               | -             | 2          |
| <b>PC8.</b> Support for vehicle integration & series implementation                                                     | 4            | 4               | -             | 2          |
| <b>PC9.</b> Participate quality meetings & receive BMS performance feedback                                             | 4            | 4               | -             | 2          |
| <b>PC10.</b> Understand field issues regarding BMS related to various architecture, durability or control system issues | 3            | 3               | -             | 2          |
| <b>PC11.</b> Propose & provide necessary technical resolution for the issues during deployment                          | 2            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                        | <b>40</b>    | <b>40</b>       | -             | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                         |
|----------------------------|---------------------------------------------------------|
| <b>NOS Code</b>            | ASC/N8338                                               |
| <b>NOS Name</b>            | Support manager to execute implementation of BMS system |
| <b>Sector</b>              | Automotive                                              |
| <b>Sub-Sector</b>          | Research & Development                                  |
| <b>Occupation</b>          | Automotive Product Development                          |
| <b>NSQF Level</b>          | 5.5                                                     |
| <b>Credits</b>             | 5                                                       |
| <b>Version</b>             | 1.0                                                     |
| <b>Last Reviewed Date</b>  | 28/02/2023                                              |
| <b>Next Review Date</b>    | 28/02/2026                                              |
| <b>NSQC Clearance Date</b> | 28/02/2023                                              |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

### Minimum Aggregate Passing % at QP Level : 70

## Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

### Compulsory NOS

| National Occupational Standards                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|---------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| ASC/N9810.Manage work and resources (Manufacturing)                                   | 50           | 30              | -             | 20         | 100         | 15         |
| DGT/VSQ/N0103.Employability Skills (90 Hours)                                         | 20           | 30              | -             | -          | 50          | 10         |
| ASC/N8335.Review the targeted design architecture of EV                               | 40           | 40              | 0             | 20         | 100         | 15         |
| ASC/N8336.Develop prominent options of BMS architecture, infrastructure and solutions | 40           | 40              | 0             | 20         | 100         | 20         |
| ASC/N8337.Conduct simulation for verification & validate of various architectures     | 40           | 40              | 0             | 20         | 100         | 20         |
| ASC/N8338.Support manager to execute implementation of BMS system                     | 40           | 40              | 0             | 20         | 100         | 20         |
| <b>Total</b>                                                                          | <b>230</b>   | <b>220</b>      | <b>-</b>      | <b>100</b> | <b>550</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |